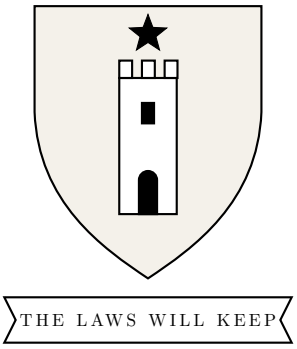
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ON A LOWER BOUND FOR THE LANDING OF DEFLECTED TRANSLATION

By Harding

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ON A LOWER BOUND FOR THE LANDING OF DEFLECTED TRANSLATION

By Harding*

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Summary. Deflected translation projects the caster into the sea of mana and returns him at another place, and no one has ever reckoned its landing exactly. This paper first asks what it means to reckon the landing beforehand: only what can be had before the translation is cast may be used, and the condition of the sea at the point of entry can be learned only by exchanging mana with it. Under a bridging premise, that the exchanges at the point of entry also obey the relation of raising and lowering of Selby and Ormes, the place and the current of the sea at one point cannot both have definite values, the product of their spreads being not less than half a wisp; the part of the landing along the current depends on the place at the point of entry, the part across it on the current there, and both are magnified by the passage through the sea. It is proved that, under the tested conditions of the model, for a given planned reach and phase of the tide, every landing reckoned only from what can be had before the casting has a mean error not less than B, namely $\sqrt{0.92\varepsilon ab}$, where ε is the wisp, a and b are the two magnifications of the passage, and 0.92 is an allowance for the terms the model neglects. The premises of the model are taken from more than seven thousand nine hundred old traces in the Register of Translation Traces of the White Tower, and tested independently on more than two thousand three hundred signed traces formed since F.E. 331. The latter half of the paper gives a method of locating that approaches the bound, namely a balanced sounding of the sea before the casting; in a hundred and eighty-four drills by twelve law-mages, the ratio of the mean error to B did not exceed 1.2 in any band. The distribution of landings is also given. The formal proof of the theorem has more than one million and forty thousand derivations, checked one by one by a verifier independent of the author and of the inference engine; what the verifier confirms is that the conclusion holds upon the model, while the model itself rests upon the test of the traces.

Introduction

Severin's Eighteenth Question asks: "Can the landing of deflected translation be reckoned beforehand?" And he goes on: can the exact landing be reckoned from what is known before the casting alone; and if it cannot, how small at least can the error be made [1]? More than a thousand years after the question was put, no one has yet reckoned it exactly. In the classical age some reckoned from the phase of

*White Tower University, White Tower City. The formal text of the proof is sealed by signature; its digest is printed at the end of this paper.

the tide and the place of setting out; the landings so reckoned commonly missed by more than three tenths of the reach, and sometimes fell far off [2].

To answer the question one must first make clear what "reckoning beforehand" means. Whether a landing can be reckoned depends on what one holds when reckoning. Deflected translation first projects the caster into the sea of mana, and then returns him to the world at another place [3]; where he returns depends on his passage through the sea, and what the sea is like at the point of entry cannot be seen before the casting. There is only one way to learn it: to exchange mana with it and read its answer. Selby and Ormes proved that every exchange between the web and the depths is a whole number of wisps [4]. "What can be had before the casting" comes down, then, to the readings obtained by exchange with the sea before the casting, together with the place and time of setting out, the phase of the tide and the force of projection the caster intends. The question of this paper is: with these, how exactly at most can the landing be reckoned?

The answer comes in two parts. The first is a relation: the place and the current of the sea at one point — the position of the point of entry within the current of the sea, and the strength of the cross-current there — cannot both have definite values, and the product of their spreads is not less than half a wisp (§2). The second is the joining of this relation with the passage through the sea: the part of the landing along the current depends on the place, the part across it on the current, and both are magnified by the passage, the more so as the reach grows; so that the mean error of the landing has a lower bound that cannot be removed (§3, Theorem 1). The bound depends neither on the skill of the caster nor on the fineness of instruments, but only on the size of the wisp and on the passage through the sea.

The proof of the bound uses several premises of a model, and these are taken from traces. The Register of Translation Traces of the White Tower records the setting out and landing of deflected translations from C.R. 1190, and to F.E. 330 holds seven thousand nine hundred and eighty-six usable entries [5]; from these I fixed the model, and I tested it independently on two thousand three hundred and seventeen signed traces formed after signatures came into use in 331 [6] (§6). The formal proof of the theorem is long; with the help of an inference engine it was set out as more than one million and forty thousand derivations, and given to an independent verifier to check one by one (Appendix).

Section 1 defines the concepts used; §2 gives the relation of place and current; §3 proves the bound; §4 treats short and long reaches; §5 discusses some methods; §6 tests the premises; §7 gives the method of locating; §8 the distribution of landings.

1. Some concepts: the landing, the error, and what is known before casting

1.1. Setting out, landing and the cut-off By "the casting" I mean here the deflected translation alone, not the sounding or any other spell. A deflected translation has a place of setting out X and a time t, and a landing L. The cut-off is the moment the projection begins: whatever is had before it may be used, whatever is formed after it may not, and the reckoned point must be sealed at the cut-off. Before the cut-off

one can read: the place and time of setting out; the phase of the ten-year tide and of the daily rise and fall; the bearing and brightness of the twelve magic moons, which are the reflection of the sea of mana in the sky [7]; the force of projection the caster intends; and any readings of soundings made before the cut-off (§1.3). The actual force of projection recorded afterwards in the trace of setting out is not knowledge had before the casting, since the projection may deepen unintentionally at entry (§8). The greater the force, the farther the landing as a rule; but neither its direction nor its distance is fixed by the caster [3].

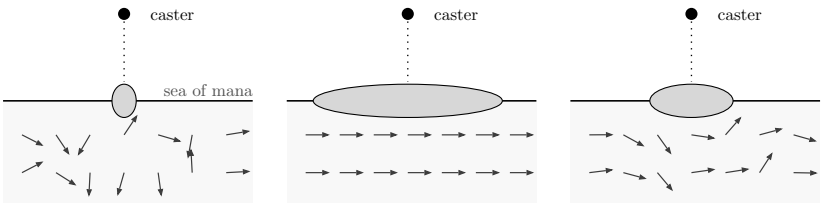
1.2. The reckoned point, the error and the reach Let I be all that can be had before the cut-off, and f any means of reckoning a point from I; call $f(I)$ the reckoned point. The error $\delta = L - f(I)$ is a vector in the plane. By the mean error I mean the square root of the mean of the square of $|\delta|$; if only the mean of the distance is taken, the conclusion is the same, with a smaller constant (§3.4). The planned reach R is the distance of the mean landing reckoned by the model from the intended force of projection; it does not change with the means of reckoning being compared. The theorem compares all means of reckoning at a given R and phase of the tide; the degenerate case $R = 0$ is not treated.

1.3. Sounding the sea To learn the condition of the sea before casting, a low-ring sounding of the sea may be cast at the place of setting out: the sounding reaches down to the sea of mana by the caster's own mana, exchanges some wisps with it, and reads back the answer of the sea at that point. This is the only means of exchange with the point of entry before the cut-off; the phase of the tide and the moons read the general state of the sea, not its condition at the point of entry.

1.4. The current In the old traces, landings from one place in one season lean, on the whole, to one side, and that side agrees with the general state of the sea shown by the moons at the time. I call this direction the current, and divide the error into a part along the current, $\delta_{\parallel}$, and a part across it, $\delta_{\perp}$.

2. Place and current

2.1. A thought Suppose the caster wishes to read exactly, before setting out, the position of the point of entry within the current. He must make the sounding exchange with a very small region only, to know which region it is; but the smaller the region of exchange, the more wisps are needed, and every wisp pushes the sea there a little, so that the cross-current there is thrown into disorder. Conversely, to read the cross-current exactly, the sounding must exchange slowly with a wide stretch of the sea, and then it cannot say which point it was. The more exactly the one is read, the more the other is disordered (Fig. 1).



(a) narrow sounding: place read well, current disordered

(b) wide sounding: current read well, place uncertain

(c) balanced sounding: the two parts in balance

FIG. 1. The thought of sounding the sea: at the place of setting out, a sounding that exchanges with a small region reads the place well and throws the current into disorder; one that exchanges with a wide region reads the current well and cannot tell the place; a balanced sounding keeps the two parts in balance.

2.2. Derivation from the relation of raising and lowering By the general rules since Elworthy, the exchange between the web and the depths is described by two operations, raising U and lowering D [8]; Selby and Ormes proved that the difference of their two orders is not nought,

$$DU - UD = \varepsilon \tag{1}$$

ε being the wisp [4]. What Selby and Ormes proved concerns the exchange between the web and the depths, and their reading of one opening and closing as one wisp is a further premise of their own. This paper needs a further, bridging premise (Premise 2 of §3): that the exchanges of the sea at the point of entry with the sounding and with the return also obey (1), with the same ε. Under this premise I call the condition of the sea at the point of entry its sea-state, written, added and measured in strength by the rules for web-states, and I consider only sea-states in which place and current have finite second moments. Take

$$S = \frac{U + D}{\sqrt{2}}, \quad V = A\frac{D - U}{\sqrt{2}} \tag{2}$$

where A is Elworthy's warp, which may be interchanged with U and D and whose square under successive application is the negative of the identity operation [8]. By (1), $SV - VS = -\varepsilon A$, whose strength in every sea-state is ε (Lemma 2.1 of the formal text). By Elworthy's rules of duals and strength, for every sea-state the sea can be in,

$$\sigma_S \cdot \sigma_V \geq \frac{\varepsilon}{2} \tag{3}$$

where σ denotes the spread of a quantity in that sea-state, that is, the square root of the mean square of its difference from its mean. This is the relation that the thought of §2.1 calls for, and it is stronger than that thought: it does not say that the sounding is clumsy, but that in every sea-state the sea can be in, place and current are never both definite. Sea-states in which (3) holds with equality exist; in them place and current are both distributed normally, and their spreads may be traded one against the other.

2.3. After sounding It must also be made clear that (3) still holds after the sea has been sounded.

Lemma 1. *However many soundings are made, and however each is chosen after the readings of the last, once the readings are had the conditional sea-state at the point of entry is still a sea-state, and satisfies (3).*

The conditional sea-state here is the sea-state renewed, according to the rule of reading, by the shares read. The conditions of Lemma 1 are that every sounding exchanges with the sea only through raising and lowering, and that the readings follow the rule of reading of Selby and Ormes; its proof is in the formal text, where it takes up most of §2. By Lemma 1, at the cut-off, whatever has been done before, the point of entry is in a sea-state satisfying (3).

3. The lower bound

3.1. The model The bound rests on four premises.

(Premise 1) The mechanism of return turns the sea-state at the point of entry at the cut-off into a distribution of landings: given I, the part along the current, $\delta_{\parallel}$, has the same distribution as $a\big(S-\hat{S}\big)$ in that conditional sea-state, and the part across it, $\delta_{\perp}$, the same as $b\big(V-\hat{V}\big)$:

$$\delta_{\parallel} \sim a\big(S-\hat{S}\big), \quad \delta_{\perp} \sim b\big(V-\hat{V}\big) \tag{4}$$

$\hat{S}$ and $\hat{V}$ are any estimates made from I; a and b are the two magnifications of the passage, both positive, which grow with the planned reach. The joint distribution of the two parts is not used; if the return, reading both parts at once, adds noise, it only makes the error greater. Terms of the second order and higher are neglected in (4); within the reaches covered by the first batch, the neglected terms together with their cross terms with the linear ones reduce the mean square error by at most eight parts in a hundred, so that η does not exceed 0.08 (§6).

(Premise 2) The exchanges at the point of entry obey (1), which is the bridging premise of §2.2; hence (3) and Lemma 1.

(Premise 3) Whatever is known before the cut-off about the point of entry comes by exchange with the sea; there is no other way.

(Premise 4) The dependence of a and b on the planned reach R is fixed by the traces of pairs that set out together (§6); at the ordinary phase of the tide

$$\sqrt{\varepsilon ab} = 0.438\sqrt{R}e^{R/600} \tag{5}$$

R and the left side both being in miles, for R not exceeding four hundred miles. At short reaches it grows as the square root of the reach, like a drop of ink spreading in water; at long reaches it gains a further factor growing exponentially with the reach. At the ebb there is another such formula, with 690 miles in place of the 600 in the exponent (§5.1).

3.2. The theorem

THEOREM 1. *Under Premises 1 to 4, for a given planned reach R (not exceeding four hundred miles) and phase of the tide, for every means of sounding the sea before the casting and every means $\mathbf{f}$ of reckoning the point,*

$$\text{mean error} \geq B = \sqrt{(1-\eta)\varepsilon ab} \tag{6}$$

At the ordinary phase of the tide, $B(R) = 0.42\sqrt{R}e^{R/600}$ miles.

Proof. First fix I. By Lemma 1 the point of entry is in a conditional sea-state satisfying (3). The mean square of the difference of a quantity from any estimate is not less than the square of its spread, so that by (4), the neglected terms set aside for the moment,

$$E\big(|\delta|^2 \mid I\big) \geq a^2\sigma_S^2 + b^2\sigma_V^2 \geq 2ab\sigma_S\sigma_V \geq \varepsilon ab \tag{7}$$

the middle step being that the sum of two squares is not less than twice their product. Taking the neglected terms into account removes at most a fraction η more, so that the conditional mean square error is not less than $(1-\eta)\varepsilon ab$. With R and the phase of the tide fixed, a and b do not vary with I, and averaging over I the whole mean square error is not less than this either; if reaches and phases of the tide are pooled, the bound is the square root of the mean of $(1-\eta)\varepsilon ab$ weighted by their proportions.

The proof is short because the difficulties have all been put into the premises. What the formal text must prove is Lemma 1; the estimate of the neglected terms and their cross terms; that the landing lies in three dimensions, the vertical part being counted too; and that (3) itself follows from (1) by Elworthy's rules.

3.3. What the bound means (6) does not say that the present methods are not good enough, but that no method can be better: so long as the premises of the model hold, no sounding however fine and no engine however great can reckon the landing with a mean error less than B. In the ideal linear model in two dimensions, with η nought and no vertical error, the bound can be attained: the reckoned point must be the conditional mean, the sea-state must be one in which (3) holds with equality, and $a\sigma_S = b\sigma_V$, that is, the parts along and across the current must be in balance. Actual spells can only approach it. Table 1 gives the bound at several reaches at the ordinary phase of the tide.

TABLE 1. The lower bound of the mean error at several reaches (ordinary phase of the tide)

Planned reach R (miles)	*5*	*20*	*50*	*100*	*200*	*400*
Bound B (miles)	0.95	1.9	3.2	5.0	8.3	16.4

3.4. The mean of the distance If the mean error is taken as the mean of the distance only, another form of (3) is needed: that the sum of the entropies of the distributions of place and current is not less than $\ln(\pi e\varepsilon)$. This does not follow from (3), and the formal text proves it separately. From it, and from the fact that the mean deviation of a distribution is not less than $e^{h-1}/2$, where h is its entropy, the mean distance is found to be not less than $0.76B$; the balanced circular normal distribution gives

$\sqrt{\pi}/2$ times B, that is, $0.89B$. Which number between these is the exact bound I have not determined.

4. Short reaches and long

In the manuscripts of the classical age several law-mages claim to have landed without the least error [2]. The reaches they record are none more than a mile or two; by (6) the bound there is about half a mile, and with the measures of that time, to come down in the wood or on the hill one had in mind was to land without the least error. As the reach grows the bound grows as $\sqrt{R}$, and at some tens of miles the landing is out by some miles; at some hundreds of miles the exponential factor takes effect, and the bound grows much faster than the square root of the reach. When Severin says "sometimes it falls far off", he speaks most likely of deflections of long reach.

This shows also why deflected translation serves for escape: at any but a short reach the caster himself cannot reckon where he will land, and the pursuer, who has only what can be seen from outside, still less.

5. Discussion of some methods

5.1. Choosing the tide At the ebb the sea is calmer; on the first batch the 600 miles in the exponent of (5) becomes 690 miles, and at a reach of four hundred miles the bound is some eight parts in a hundred smaller. Those who set out on the ebb therefore have a smaller bound at long reach; the theorem holds separately for each phase of the tide, and choosing the tide does not alter it, but alters a and b.

5.2. Sounding many times If one sounding is not enough, several do not lower the bound: by Lemma 1 the sea-state after sounding still satisfies (3). Sounding many times steadies the readings, but cannot pass (6).

5.3. Reading the moons The moons are the reflection of the sea, and show its general state and the direction of its current, but not the place and current at the point of entry. Reading the moons fixes the direction of §1.4, but cannot bring the error below the bound.

5.4. Borrowing an array It has been proposed to pin the point of entry with an array, so as to make the place definite. Borrowing an array changes the mechanism of entry, and whether it still satisfies Premise 1 must first be checked. By (3), the harder the place is pinned the more the current is disordered; the first batch contains nineteen traces of setting out from an array, and taken as a group their mean square error is four tenths greater than that of a group without arrays matched in reach and phase of the tide.

6. Test of the premises

6.1. Three batches of traces The model was fixed on the first batch, tested on the second, and the method of locating tested on the third (Table 2). The first batch consists of the entries of the Register of Translation Traces of the White Tower from C.R. 1190 to F.E. 330 [5]; of the 11,214 entries, only those whose traces of setting out and of landing are both whole and whose times agree were taken, the 1,402 whose landings were never found, the 1,105 with damaged traces and the 721 whose

times disagree being set aside. The second consists of the entries from 331 to 356, all signed, every signature checked [6]; of the 2,618 entries, the 193 whose landings were never found and the 108 with damaged traces were set aside. The rules of choice were fixed before any landing was looked at. The model was completed in the Month of the Winter Lyre, 351, and sealed by signature, and the landings of the second batch were drawn out for comparison only afterwards. The third batch is the drills of 352 to 357, kept in a separate book and not among the second batch of the Register, described in §7. By the rule followed since Darby, the model was fixed on the first batch alone, the second and third taking no part [9]. Landings never found are mostly far ones, so that the conclusions of this paper concern deflected translations whose landings were found, and the proportion of far landings may be too low.

TABLE 2. The three batches of traces

Batch	*Years*	*Entries*	*Taken*	*Of which pairs*	*Use*
First	C.R. 1190 – F.E. 330	11,214	7,986	643 pairs	Fixing the model
Second	F.E. 331–356, signed	2,618	2,317	171 pairs	Testing the premises
Third	F.E. 352–357, drills	184	184	—	Testing the method of locating

6.2. Magnification A pair is two casters setting out from one place within a few paces or a few breaths of each other. What the two know before the cut-off is almost the same, but their points of entry differ. For pairs setting out a few paces apart at almost the same moment, the places at their points of entry differ by about the distance between their places of setting out, and how far apart their landings fall along the current shows a; for pairs setting out from one place a few breaths apart, the difference of the currents at their points of entry is estimated from the change of the general current shown by the moons over those breaths, the local differences averaging out over many pairs, and how far apart their landings fall across the current shows b. Of the 643 pairs of the first batch, 291 are of the former kind and 352 of the latter. From them a and b are fixed and combined into (5): the coefficient 0.438 ± 0.011, and the 600 ± 40 miles in the exponent; the ratio of a to b is 1.12 ± 0.09, and changes little with the reach. The 171 pairs of the second batch, fixing them again, give 0.427 ± 0.019 and 585 ± 70 miles, agreeing within the errors. The first batch spans the Great Tide, and the second the remaking of the standards by the Seventh Division in 342; reaches are measured in miles and are untouched by the change of the units of mana, and when the two parts of the first batch before and after the Great Tide are fitted separately the coefficients also agree within the errors.

6.3. The neglected terms In (4) terms of the second order and higher have been neglected. Within the reaches covered by the first batch, taking the neglected terms and their cross terms with the linear ones into account changes the mean square

error by not more than eight parts in a hundred, so that $\eta = 0.08$ is taken; so also on the second batch. Traces with a reach beyond four hundred miles are too few, and the model does not extend there.

6.4. Forecast from outward knowledge Without sounding the sea, from the place and time of setting out, the phase of the tide and the moons only, the model can still reckon a landing, with an error much greater than the bound. Compared with the second batch (Table 3, Fig. 2), the observed errors in all six bands are four to six parts in a hundred above those forecast by the model, all in the same direction; most likely because, since signatures came into use, landing traces are more easily found and more of the far landings have been recorded, while the model, fixed on the first batch, has too few far landings. Of the four premises, Premises 1 and 4 are tested directly by the pairs and by the second batch; Premise 2 and Lemma 1 are tested only indirectly, in that the errors of the third batch fall below the bound in no band; Premise 3 is an assumption of this paper.

Table 3. The second batch: mean errors of the several methods (miles)

Planned reach (miles)	*Entries*	*Bound B*	*Old method*	*Outward knowledge, forecast*	*Outward knowledge, observed*
below 10	412	1.02	2.6	2.2	2.3
10–30	538	1.84	7.0	4.2	4.4
30–60	497	2.96	16.0	6.9	7.2
60–120	446	4.43	31.2	10.6	11.0
120–250	318	7.30	62.1	18.3	19.1
250–400	106	12.40	112.3	32.2	34.0

The “old method” is that of the law-mages since the classical age: to take the old traces of setting out from the same place in the same season, and use the mean of their landings as the reckoned point [2]. The bound B of each band is the square root of the mean of the squares of $B(R)$ over the traces of that band; the reach and B of every trace are deposited. Every column includes the far landings.

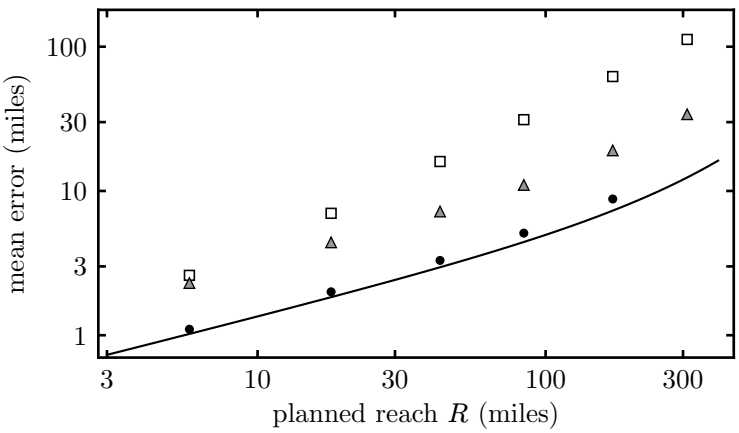


FIG. 2. Mean error against planned reach: the curve is the bound $B(R)$; squares, the old method; triangles, the second batch with outward knowledge only; circles, the sounding and locating of the third batch.

7. The method of locating

By locating I mean reckoning before the casting where the landing will most likely be, not sending the caster to a chosen place. Section 3.3 shows how the bound may be approached: the parts of the error along and across the current must be in balance, and the sea must therefore be sounded neither for the place alone nor for the current alone, but in balance according to the ratio of a to b (§6.2). The size of the region with which the sounding exchanges is taken such that the spreads of the parts along and across the current are equal; once the readings are had, the reckoned point is the conditional mean reckoned by the model, sealed at the cut-off.

The third batch consists of the drills made by this method by twelve law-mages of the White Tower between 352 and 357, a hundred and eighty-four in all. They set out far from cities and sea-lanes, with a ship standing by each time. The results are in Table 4: in every band the ratio of the mean error to B does not exceed 1.2, the band of shortest reach coming nearest, at about 1.06; in no band is the mean below the bound. No drill had a reach beyond two hundred and fifty miles.

TABLE 4. The third batch: mean errors of sounding and locating (miles)

Planned reach (miles)	*Drills*	*Bound B*	*Sounding and locating*	*Ratio to B*
below 10	21	1.04	1.10	1.06
10–30	47	1.85	2.02	1.09
30–60	52	2.94	3.31	1.13
60–120	41	4.47	5.09	1.14
120–250	23	7.25	8.62	1.19

The bound of each band is reckoned as in Table 3, from the reaches of the drills of the third batch. On the same hundred and eighty-four drills a point was also reckoned by the old method and from outward knowledge only: taken together, the mean error of the old method is about 6.4 times B, that from outward knowledge about 2.4 times, and that of sounding and locating 1.13 times. Each of the twelve

law-mages made a dozen or more drills, and the drills of one caster are not wholly independent; estimated by casters, the uncertainty of the ratio for sounding and locating is about plus or minus 0.05.

8. The distribution of landings

By the model, after a balanced sounding of the sea the greater part of the error is near a circular normal distribution in two variables, with a spread of about $B/\sqrt{2}$ in every direction; besides this, about one in a hundred landings falls along the current beyond three times the bound, owing to an unintended deepening of the force of projection at entry, and of these the model gives only the proportion, not where they fall. There were two such in the third batch, both included in Table 4. With outward knowledge only, without sounding, the distribution is long along the current and short across it (Fig. 3), the spread along being about one and a half times that across, and the far landings are about four in a hundred.

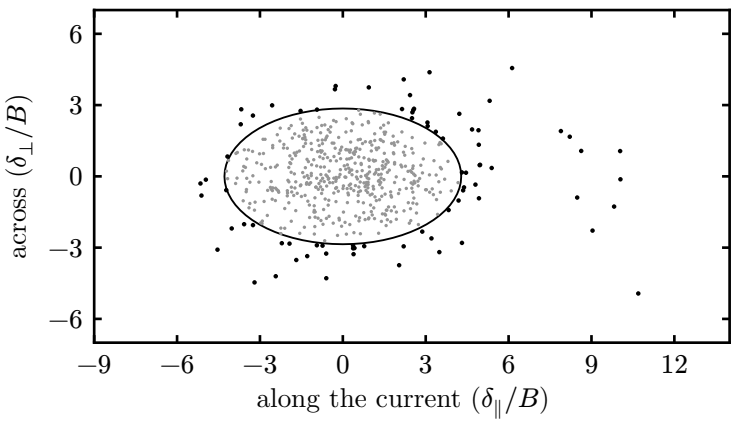


Fig. 3. The distribution of landings: the errors of the second batch with outward knowledge only, the parts along and across the current both in units of the bound B; the ellipse is the region containing nine tenths of the model's distribution; the points are 560 observed landings taken at random, those outside the ellipse in black.

These two distributions may serve those who wish to prepare for a landing, and those who wish to find one. But a distribution says only where a landing will most likely be: there are landings outside the region of nine tenths, and of the far landings there is only a proportion, no place. An order of search drawn from the distribution can follow only the estimated chances, and must not take the ellipse for a boundary.

Conclusion

Severin asked: if it cannot, how small at least can the error be made? The answer of this paper is: under the tested conditions of the model, not less than the B of (6); and by a balanced sounding of the sea the drills came within 1.2 times of it. The bound rests on the premises of the model, and the premises on the test of the traces: should anyone find hereafter a means of knowing the condition of the sea without exchange, Premise 3 would fail, and the bound would have to be examined anew.

Why the passage through the sea magnifies in this way, I do not discuss. Formula (5) is only the form fixed by the traces; why it spreads at short reach and grows like a rolling snowball at long reach is for those who study the sea of mana to answer.

Acknowledgements. I first read this question in the library of White Tower University when I was fifteen. In the sixty-five years since, the trace-officers of the White Tower copied out for me more than ten thousand settings out and landings from the Register; twelve law-mages took the risk of the drills of the third batch; the engine rooms of White Tower University and of the University of Kanaich ran the formal text for me; and those at the University of Kanaich who wrote the verifier had never seen my derivations before I handed in the formal text.

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3. The plane-scholars of Chaen, The Register of Planes, revised edition, C.R. c. 1480; the entry on deflected translation.
4. Selby and Ormes, "On the least share in the exchange of mana", Transactions of the Silver Society, Series I (F.E. 338), no. 3.
5. The White Tower, The Register of Translation Traces, from C.R. 1190.
6. Oxley, "Signatures of traces", Runes and Formulae 91 (F.E. 331).
7. Aldridge, "On the tides", Transactions of the Silver Society (C.R. 1101).
8. Elworthy, "Nomological algebra", Annals of the White Tower Society (C.R. 1061).
9. Darby, The inductive formula, Omnium College, c. F.E. 265.
10. Kanaich Institute of Technology and the Great Library, "The first inference engine", Bulletin of the Great Library (F.E. 353), no. 4.
11. The Academic Council of the Silver Society, rule on the checking of results of inference engines, c. F.E. 355.
12. Finley, "Mechanically checkable proofs of safety for finite-state control formulae", Annals of the Laws (F.E. 213).

Appendix. The record of the formal text and its verification

Under the rule of about 355 [11], this appendix records how the formal text was written, who verified it, and what was verified.

The formal text is written in the notation of the verifier of the University of Kanaich; since Finley, verifiers of this line accept only derivations made step by step in their notation [12]. I first wrote the skeleton of Theorem 1: nine hundred and twelve lemmas, with the statement of each and the outline of its proof; the inference engine [10] expanded the proof of every lemma into derivations for the verifier, the manner of expansion and the versions of the formulae used being separately recorded. Expanded, the text has 1,046,233 derivations, of which the derivation of (3) from (1), together with Lemma 1, takes 386,402; the estimate of the terms neglected in (4) and their cross terms, 302,155; the inequality of entropies and the bound on the mean distance of §3.4, 147,318; and the remaining 210,358 are notation, definitions and general lemmas. Whatever derivations expanded by the engine the verifier did not accept were sent back to me, the skeleton rewritten, and expanded again by the engine; this went back and forth more than two hundred

times. The formal text carries a list of the lemmas, a list of the axioms and definitions used, and a table matching each formula of this paper with its proposition in the formal text.

The verifier is the fourth edition of that of the University of Kanaich, written by a group there, and unconnected with the engine or with me. The formal text was verified once in each of the engine rooms of White Tower University and of the University of Kanaich, taking forty-one days at each, and passed wholly at both; both rooms used the same edition of the verifier, so that the two runs exclude only chance faults of the machines, and are not a verification by two verifiers differently written. The formal text is sealed by signature [6], and its digest is printed below, so that anyone may be sure that what he verifies is the same text.

What the verifier confirms is that, upon Premises 1 to 4 as written in the formal text, Theorem 1 holds. It does not confirm that the premises agree with the facts; that is the business of §6. Nor does it confirm, on the referees' behalf, that the propositions of the formal text say what this paper says; for that, see the table of matching just mentioned.

The digest of the formal text, by Oxley's method of compression, its first sixty figures: 09536 82363 99170 68044 03014 88246 15428 84283 72112 33297 03059 37497.

Note added in proof. After this paper had been set in proof, Ormes of Tumen University read §2 and wrote to point out that the thought of §2.1 may lead one to suppose that the bound arises because the sounding pushes the sea, as though a lighter sounding might avoid it. In truth (3) holds for every sea-state the sea can be in, sounded or not; sounding is only the sole means of learning the condition of the sea, and does not make the bound. I agree. Section 2.1 is only a way of seeing it; the proof of §3 does not depend upon it.